Sieve and Sage: Efficient Distraction Filtering for Reliable RALM Abstention

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Abstract

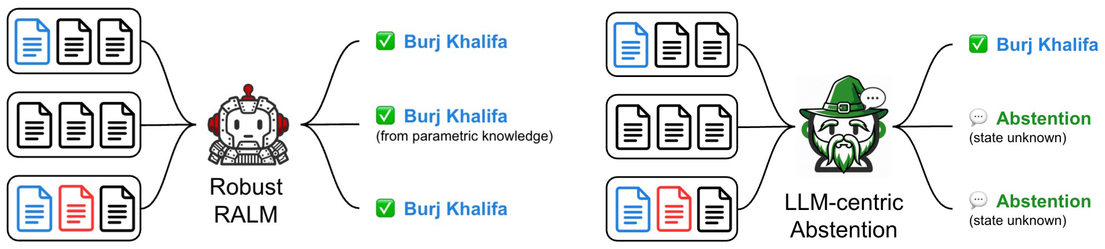
Handling different retrieval failures with monolithic LLM is costly.

Instead, we propose a two-stage framework: **Sieve and Sage**.
- **Sieve** (lightweight classifier) detects distracting documents and triggers early abstention before invoking LLM.
- **Sage** (LLM) generates a grounded answer or abstains when evidence is insufficient.

Sieve and Sage framework has two advantages.
- It achieves up to 1.9x faster inference speedup and improves system accuracy by 69.4%p. (vs. DTA-chatQA / vanilla LLM)
- It enables state-aware follow-up strategies that target the cause of abstention, rather than treating every abstention the same way.

Background

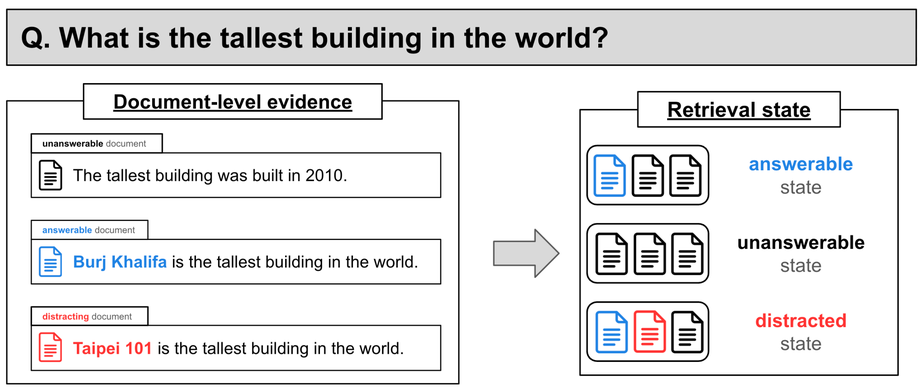
- Prior works focus on Robust RALM under noisy retrieval, however in expert domains, **abstention can be more valuable than a correct but ungrounded answer.**
- Prior RALM abstention methods are largely **LLM-centric, incurring substantial computational costs**. (e.g. post-training[1], self-consistency[2])



Problem Formulation

We categorize retrieved document sets into **three retrieval states** based on their document-level evidence composition:

- **Answerable**: Supporting evidence is present without distracting evidence.
- **Unanswerable**: The required evidence is absent.
- **Distracted**: Supporting evidence coexists with distracting evidence such as conflicting, negated, or adversarial documents.

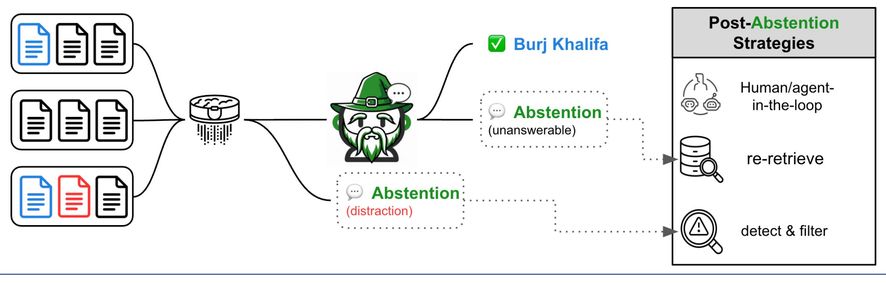


Methodology

Sieve and Sage framework

- **Sieve** is a lightweight classifier to detect distracting evidence and trigger early abstention.
- **Sage** answers grounded queries or abstains when evidence is insufficient.
- **State-specific abstention signals** guide different follow-up strategies.

- **Training objective**:
  - (Sieve) standard BCE b/w answerable and conflict retrieval states
  - (Sage) standard SFT for grounded answers and unanswerable abstention



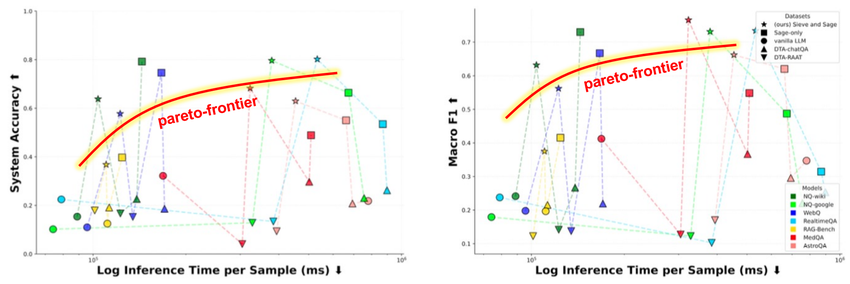
Experiments & Results

Evaluation setup

- Datasets: Five general-domain and three expert-domain QA benchmarks
- Model backbone
  - (Sieve) Longformer-base-4096 adapted with < 200 domain documents
  - (Sage) Llama3-8B-Instruct or domain-specific variants
- Metrics: System Accuracy, Macro-F1, Inference latency
- Baselines: Vanilla LLM, Sage-only(LLM ft. only) , DTA-chatQA[3], DTA-RAAT[3]

Main Results: Accuracy and Efficiency

- Across diverse datasets, our framework (★) consistently lies on the Pareto frontier with respect to both system accuracy and macro-F1.
- It achieves up to a 1.9x inference speedup and improves system accuracy by 69.4%p. (vs. DTA-chatQA / vanilla LLM)



(a) Inference time vs. System accuracy (b) Inference time vs. Macro F1

Abstention State-Aware Follow-up Strategies

- Sieve and Sage enables **effective targeted recovery strategies after abstention** by distinguishing unanswerable and distracting evidence

Dataset	Strategy	Accuracy
NQ-google	No Recovery	0.247
	+ Distraction Filtering	0.867
	+ Agent-in-the-loop	0.716
RealtimeQA	No Recovery	0.380
	+ Distraction Filtering	0.943
	+ Agent-in-the-loop	0.774

Component Analysis

- Fine-tuning LLM(Sage) or classifier (Sieve) alone does not yield the best performance.
- Sieve and Sage achieves the highest performance by separating distraction detection from grounded answering and abstention.

Dataset	Configuration	System Acc.	Macro-F1
NQ-google	Vanilla	0.102	0.179
	+ Sage ft.	0.664	0.487
	+ Sieve ft.	0.769	0.695
	+ both ft.	0.796	0.731
RealtimeQA	Vanilla	0.225	0.238
	+ Sage ft.	0.535	0.314
	+ Sieve ft.	0.791	0.728
	+ both ft.	0.802	0.734

- **Why binary training for Sieve?**
  The t-SNE plot shows **clearer separation of distracting cases** than three-way training, including unseen negated and adversarial examples.



(c) Binary: S_{ans} vs. S_{conflict} (d) Three-way: S_{ans}, S_{unans}, and S_{conflict}

Conclusion & Future work

Conclusion
- We distinguish three retrieval states: answerable, unanswerable, and distracted.
- Sieve and Sage framework enables efficient abstention: Sieve detects distractions early, while Sage answers or abstains when evidence is missing.
- State-specific abstention can guide targeted follow-up actions.

Future work
- Validate the framework on naturally occurring retrieval failures beyond simulated noise.
- Reduce the need for labeled target-domain examples.
- Calibrate abstention to reduce incorrect answers without unnecessarily sacrificing coverage.

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References

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2 Joren, Hailey, et al. "Sufficient context: A new lens on retrieval augmented generation systems.
3 Sun, Xin, et al. "Divide-then-align: Honest alignment based on the knowledge boundary of rag

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